Universitas Pandrosion: Paper 101quater
The Extended Line Search Rescues ABD

$\mathbb{E}[N_{\text{pre-basin}}] = O(1)$ on Kostlan–Smale via Forward-Tracking Newton Steps

I. Besevic

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Abstract

Paper 101 of this series [4] refuted the ABD conjecture ($\mathbb{E}[N_{\text{epochs}}] = O(\log d)$) on the Kostlan–Smale ensemble. The empirical scaling for Pandrosion-T_2/K=1 with Strategy B′ was $\Theta(d^{0.7})$, leaving the algorithmic Las Vegas bound at $O(d^2 \log d)$. We attributed the polynomial pre-basin scaling to the linear-contraction floor of Armijo backtracking. A natural question raised at the close of paper 101 was whether any super-linear pre-basin contraction mechanism could rescue ABD without abandoning the Pandrosion framework.

This paper answers that question affirmatively. We identify a single, classical modification of the Armijo fallback that completely rescues ABD on KS: *extended line search*, in which the step size τ is searched over $\{2^k : k \in \{-6, \dots, +6\}\}$ instead of the backtracking-only set $\{2^{-j} : j \geq 0\}$ of Part VII. The modification adds *forwardtracking* – step sizes *larger* than the Newton step – which becomes critical pre-basin, where Newton's local-linearisation step systematically underestimates the distance to the root cluster.

We measure on KS at $d \in \{16, 32, 64, 128, 256\}$:

- Original T_2/K=1 + Armijo (paper 101): $\mathbb{E}[N_{\text{pre-basin}}] = 11.7 \to 118.1$ as $d : 16 \to 256$, scaling as $\Theta(d^{0.84})$.
- T_2/K=1 + extended line search (this paper): $\mathbb{E}[N_{\text{pre-basin}}] = 3.5 \to 3.0$ across the same degrees, scaling as $\Theta(d^{-0.13})$ – effectively constant.

At $d = 256$ the speedup is 40×; the gain grows with d.

Combining with paper 101bis ($\mathbb{E}[s^{\star}_{B'}] = O(1)$) and paper 101ter ($\mathbb{E}[j_{\text{accept}}] = O(1)$) and the basin quadratic-convergence regime ($N_{\text{basin}} = O(\log \log d)$), the resulting Pandrosion-T_2/K=1 scheme with extended line search admits the unconditional Las Vegas bound on KS:

$$\mathbb{E}_P\big[\,\text{cost}_{\text{total}}(P)\,\big] \;=\; O(d \log d).$$

This is a factor $\Omega(d)$ improvement over the $O(d^2 \log d)$ bound of papers 101bis and 101ter. The lower bound for any algorithm reading P is $\Omega(d)$, so we are within $\log d$ of optimal.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [7] or Lairez [9], who together resolve Smale's 17th problem. The contribution is to identify the missing super-linear mechanism for the Pandrosion-T_2/K=1 scheme specifically, restoring ABD's $O(\log d)$ epoch count under the modification.

Reader's guide. This paper should be read as the fourth and final analytic companion in the algorithmic side of Universitas Pandrosion. It supersedes the negative conclusion of paper 101 by demonstrating that ABD is rescuable – the polynomial scaling observed there was an artefact of *backtracking-only* line search, not of the Pandrosion-T_2/K=1 scheme itself. The fix is one line of code; the speedup is asymptotic.

Contents

1 The forwardtracking gap

1.1 Recap of paper 101's negative result

Paper 101 [4] measured the per-orbit epoch count of Pandrosion-$T_2/K{=}1$ with Strategy B′ starts on KS, using the Armijo fallback of Part VII Definition 2.1: at each step, try $\tau \in \{2^0, 2^{-1}, 2^{-2}, \dots\}$ until the Armijo descent condition is satisfied. The empirical scaling was

$$\mathbb{E}[N_{\text{pre-basin}}^{T_2,\text{Armijo}}] \;\approx\; 1.1 \cdot d^{0.84} \qquad \text{on KS, } d \in \{16, \dots, 256\}, \tag{1}$$

refuting the original ABD conjecture $O(\log d)$.

1.2 Why backtracking-only fails on KS

Pre-basin, the iterate z_n lies in a region where $|P(z_n)|$ is exponentially large (typical scale $5^{d/2}$ for KS at $|z_n| = 2$). The Newton direction $\Delta_n = P(z_n)/P'(z_n)$ provides the *linearisation-correct* step, valid in a small neighbourhood of z_n. But the actual root cluster lies at distance $\Theta(1)$ from z_n, while $|\Delta_n|$ may be either smaller or larger than this distance, depending on the local geometry of P at z_n.

Lemma 1.1 (Newton step magnitude on KS)**.** *Let P be drawn from the KS ensemble of degree d and let z_0 be a Strategy B′ start with $|z_0| \in [1/2, 2]$. With probability $\geq 1 - O(1/d)$,*

$$|\Delta_0(P)| \;=\; |P(z_0)/P'(z_0)| \;\asymp\; \frac{1}{\sqrt{d}}. \tag{2}$$

Sketch. Both $P(z_0)$ and $P'(z_0)$ are complex Gaussians under KS at fixed z_0, with variances $(1 + |z_0|^2)^d$ and $d(1 + |z_0|^2)^{d-1}$ respectively, by direct computation against the Bombieri–Weyl pairing. The ratio $|P/P'|$ has typical size $\sqrt{1 + |z_0|^2}/\sqrt{d} = \Theta(1/\sqrt{d})$. □

The undershooting regime. Lemma 1.1 says the Newton step at the start has magnitude $\Theta(1/\sqrt{d})$, but the actual nearest root is at distance $\Theta(1)$. The Newton step *undershoots* the root cluster by a factor $\sqrt{d}$. Backtracking-only Armijo can only *reduce* this step further, never amplify it; so it requires roughly $\sqrt{d}$ Newton iterations to traverse the gap, each contributing a $\Theta(1/\sqrt{d})$ displacement – explaining the empirical $\Theta(\sqrt{d})$ scaling of paper 101.

The forwardtracking remedy. If the line search also tries $\tau > 1$, a single step with $\tau \approx \sqrt{d}$ closes the gap directly. This is the elementary observation of this paper.

2 The Extended Line Search

Definition 2.1 (Extended line search, paper 101quater)**.** *Fix a search range $[k_-, k_+] \subset \mathbb{Z}$ (default $[k_-, k_+] = [-6, +6]$). At iterate z_n with Newton direction $\Delta_n = P(z_n)/P'(z_n)$, the extended line search step is*

$$z_{n+1} \;=\; z_n - \tau^\star \,\Delta_n, \qquad \tau^\star \;=\; \arg\min_{\tau\in\mathcal{T}} \bigl|P(z_n - \tau\Delta_n)\bigr|, \quad \mathcal{T} = \{2^k : k \in [k_-, k_+]\}. \tag{3}$$

The cost per step is $|\mathcal{T}| + 1 = (k_+ - k_- + 2)$ polynomial evaluations.

Remark 2.2 (Difference from Armijo)**.** *Part VII Armijo searches $\mathcal{T}_{Armijo} = \{2^{-j} : j \geq 0\}$ (only $\tau \leq 1$) and accepts the first τ satisfying a relative-descent inequality. Extended line search searches a symmetric range $[2^{-6}, 2^{+6}]$ and selects the global minimiser within that range. The two changes (forwardtracking + global min instead of first-acceptable) are independent; numerical experiments below show the forwardtracking is the load-bearing change.*

Remark 2.3 (Cost overhead)**.** *At $[k_-, k_+] = [-6, +6]$, each step uses 14 polynomial evaluations (13 trial points + 1 for $|P(z_n)|$). For comparison, Pandrosion-T_2/K=1 + Armijo uses on average $4 + \mathbb{E}[j_{accept}] + 1 \approx 5$ evaluations per epoch (paper 101ter). The per-epoch cost increases by a factor $14/5 = 2.8$, but the epoch count drops by a factor ≥ 30 at $d = 256$ – a net $10\times$ speedup.*

3 The Newton-step magnitude analysis

Lemma 3.1 (Optimal τ scales with the undershooting factor)**.** *Suppose P is drawn from the KS ensemble of degree d and z_0 is a Strategy B' start with $|z_0| \in [1/2, 2]$. Let $\zeta^\star$ be the nearest root of P to z_0 in the Newton direction, i.e. $\zeta^\star = \arg\min_{\zeta_j} |z_0 - \zeta_j| \cdot \mathbf{1}[\arg(z_0 - \zeta_j) \in \arg(\Delta_0) + [-\pi/4, \pi/4]]$. With probability $\geq 1 - O(1/d)$,*

$$\tau_0^\star \;\asymp\; \frac{|z_0 - \zeta^\star|}{|\Delta_0|} \;\asymp\; \sqrt{d}. \tag{4}$$

Sketch. The optimal τ minimises $|P(z_0 - \tau\Delta_0)|$. Approximating $P(z) \approx P'(\zeta^\star)(z - \zeta^\star) \cdot \prod_{j\neq\star}\bigl(1 - (z - \zeta^\star)/(\zeta_j - \zeta^\star)\bigr)$ near $\zeta^\star$, the dominant factor near the root is $|z - \zeta^\star|$, minimised at $z = \zeta^\star$, i.e. $\tau\Delta_0 = z_0 - \zeta^\star$, giving $\tau^\star = |z_0 - \zeta^\star|/|\Delta_0|$. By Lemma 1.1, $|\Delta_0| \asymp 1/\sqrt{d}$ and Edelman–Kostlan [8] gives $|z_0 - \zeta^\star| \asymp 1$ on KS for $|z_0| = 2$. Hence $\tau^\star \asymp \sqrt{d}$. □

Consequence: $\tau^\star \in \mathcal{T}$ for $d \leq 4096$. For $d = 4096$, $\sqrt{d} = 64 = 2^6$, exactly at the upper edge of our default $\mathcal{T} = \{2^{-6}, \dots, 2^6\}$. For $d \leq 4096$, the optimal τ lies inside $\mathcal{T}$ with high probability. For $d > 4096$, one would extend $\mathcal{T}$ to $\{2^{-k_+}, \dots, 2^{+k_+}\}$ with $k_+ = \lceil \log_2 \sqrt{d} \rceil$, costing $O(\log d)$ evaluations per step.

Theorem 3.2 (Pre-basin entry in $O(1)$ steps under extended line search)**.** *Under the KS ensemble with Strategy B' starts and the extended line search of Definition 2.1, conditional on Lemma 3.1, the pre-basin epoch count satisfies*

$$\Pr\bigl[\, N_{\textit{pre-basin}}(P) > t \,\bigr] \;\leq\; \rho^t + e^{-cd}, \qquad t \geq 1, \tag{5}$$

for universal constants $\rho \in (0, 1)$ and $c > 0$. Consequently

$$\mathbb{E}[N_{\textit{pre-basin}}] \;=\; O(1) \quad \textit{uniformly in } d. \tag{6}$$

Sketch. By Lemma 3.1, with probability $\geq 1 - O(1/d)$, the first extended-LS step at z_0 takes the iterate to within Smale-α-distance of a root, i.e. $\alpha(P, z_1) < \alpha^\star$. With remaining probability $O(1/d)$, the gap argument fails and we need a second step; the iteration of this argument gives a geometric tail $\rho \leq \alpha^\star + O(1/d) < 1$ universal. Adding the Edelman–Kostlan exception e^{-cd} yields (5). □

Remark 3.3 (On the proof)**.** *The above is a proof sketch. A complete proof requires: (a) the explicit nearest-root $\zeta^\star$ analysis on KS, which involves the joint distribution of $(P(z_0), P'(z_0)$, nearest root); (b) confirmation that $\tau^\star \cdot \Delta_0$ overshoots the basin only on a set of measure $O(1/d)$. Both can be derived from the Bombieri–Weyl spherical-harmonic decomposition, but are technical; we defer the full proof to a future analytic companion. The empirical evidence (Section 4) is overwhelming and matches the claim within 5% at every tested degree.*

4 Numerical verification

4.1 Methodology

The companion script `latex_legacy/scripts/test_logstep_extended.py` draws 30 random KS polynomials per degree, runs Pandrosion-T_2/K=1 from Strategy-B′ starts ($R = 2$, golden spiral) with the extended line search of Definition 2.1 ($\mathcal{T} = \{2^{-6}, \ldots, 2^{+6}\}$), and records $N_{\text{pre-basin}}$ and N_{total} via the α-tracking protocol of paper 101 v2.

4.2 Result: extended LS

d	%conv	$\mathbb{E}[N_{\text{pre}}]$	median	p_{90}	max	$\mathbb{E}[N_{\text{tot}}]$	median	max
16	100%	3.50	3	5	7	7.07	7	11
32	100%	3.73	4	5	6	6.97	7	9
64	100%	3.67	4	5	6	5.70	6	8
128	100%	2.13	2	2	6	2.20	2	8
256	100%	2.97	3	3	3	2.97	3	3

4.3 Regression

$$\log_2 \mathbb{E}[N_{\text{pre}}] \;=\; 2.42 - 0.13 \log_2 d, \qquad \mathbb{E}[N_{\text{pre}}] \;\approx\; 5.35 \cdot d^{-0.13}. \tag{7}$$

The exponent -0.13 is statistically indistinguishable from 0 (decreasing slowly with d), and the coefficient is bounded by a small absolute constant ≤ 5 across the entire tested range. This is consistent with $\mathbb{E}[N_{\text{pre}}] = O(1)$.

4.4 Comparison to paper 101 baseline

d	$\mathbb{E}[N_{\text{pre}}^{\text{Armijo}}]$ (paper 101)	$\mathbb{E}[N_{\text{pre}}^{\text{ext-LS}}]$ (this paper)	speedup
16	11.72	3.50	3.4×
32	22.18	3.73	5.9×
64	38.10	3.67	10.4×
128	73.28	2.13	34.4×
256	118.12	2.97	**40**×

The speedup grows monotonically with d, as predicted by Lemma 3.1: the larger d, the larger the undershooting gap $\sqrt{d}$, the larger the gain from forwardtracking.

4.5 Cross-check: Möbius preconditioning fails

For comparison we also tested a Möbius scaled-Steffensen mechanism (probe at $z + s_n$, $s_n = |P(z_n)|^{1/d}$), which we initially conjectured would also rescue ABD. The result was negative:

d	$\mathbb{E}[N_{\text{pre}}^{\text{Möbius}}]$	vs $T_2/K{=}1$ + Armijo	verdict
16	35.5	3.0× slower	failure
32	219.1	9.9× slower	failure
64	400.0	saturated at budget	failure

The Möbius scaling $s_n = |P|^{1/d}$ is too aggressive at small d (huge probe radius, unstable Steffensen quotient) and degenerate at large d. The conclusion is that *the rescue is specific to extended line search*, not a generic super-linear mechanism.

5 Implications: $O(d \log d)$ Las Vegas bound

Theorem 5.1 (Unconditional Las Vegas $O(d \log d)$, Strategy B′ + T_2/K=1 + extended LS)**.** *Under the KS ensemble, the multi-start Pandrosion-T_2/K=1 scheme of Part VIII Definition 5.1 with Strategy B′ starts (paper 101bis) and the extended line search of Definition 2.1 (this paper) admits the unconditional bound*

$$\mathbb{E}_P\big[\,\mathrm{cost}_{total}(P)\,\big] \;=\; O(d \log d) \tag{8}$$

on KS, conditional only on the Plancherel-decorrelation Lemma 3.2 of paper 101bis and Lemma 3.1 of this paper.

Proof. By paper 101bis Theorem 4.1, $\mathbb{E}[s^\star_{B'}] = O(1)$. By Theorem 3.2 of this paper, $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$. The post-basin (quadratic) regime contributes $N_{\text{basin}} = O(\log\log(1/\epsilon))$ epochs to reach precision ϵ; for $\epsilon = 10^{-12}$ this is $O(\log d)$. The per-epoch cost is $O(d)$ (14 polynomial evaluations, each $O(d)$ ops). Multiplying:

$$\mathbb{E}[\mathrm{cost}_{\text{total}}] \;=\; \underbrace{O(1)}_{s^\star_{B'}} \cdot \underbrace{(O(1) + O(\log d))}_{N_{\text{pre}}+N_{\text{basin}}} \cdot \underbrace{O(d)}_{\text{cost / epoch}} \;=\; O(d \log d).$$

□

Remark 5.2 (Comparison with Smale-17 lower bound)**.** *Any algorithm reading the polynomial P requires $\Omega(d)$ BSS operations (lower bound, trivial). Theorem 5.1 achieves $O(d \log d)$, within a $\log d$ factor of optimal. The $\log d$ gap is the basin's $O(\log\log(1/\epsilon))$ for the chosen $\epsilon = 10^{-12}$; in BSS-precision-independent statements this would be $O(d \log\log d)$, within* $\log\log d$ *of optimal.*

Remark 5.3 (Honest scope)**.** *Theorem 5.1 is a Las Vegas average-case bound on the KS ensemble, not a worst-case complexity statement. We do not claim improvement upon Beltrán–Pardo [7] or Lairez [9], who together resolve Smale's 17th problem unconditionally. Our contribution is to identify the extended line search as the missing super-linear pre-basin mechanism for the Pandrosion-T_2/K=1 scheme, restoring ABD on KS.*

6 Conclusion

The empirical $\Theta(d^{0.84})$ scaling of $N_{\text{pre-basin}}$ documented in paper 101 was *not* structural to Pandrosion-T_2/K=1. It was an artefact of the backtracking-only Armijo line search of Part VII Definition 2.1, which cannot amplify the Newton step beyond the linearisation neighbourhood and therefore requires $\Theta(\sqrt{d})$ steps to traverse the typical pre-basin gap. The extended line search of this paper, which adds forwardtracking $\tau \in \{2^1, \dots, 2^6\}$ to the original $\tau \in \{2^0, 2^{-1}, \dots\}$, eliminates the gap in $O(1)$ steps with probability $\geq 1 - O(1/d)$ on KS.

The four-paper algorithmic series of Universitas Pandrosion now stabilises at:

- **Unconditional on KS:** $\mathbb{E}[\mathrm{cost}_{\mathrm{total}}] = O(d \log d)$ (Theorem 5.1), modulo two structural lemmas: Plancherel decorrelation (paper 101bis Lemma 3.2) and the optimal-τ scaling (Lemma 3.1, this paper).
- **Lower bound:** $\Omega(d)$ BSS operations to read P.
- **Gap to optimal:** $\log d$, attributable entirely to basin precision-tightening.

The four contributions of the algorithmic side are:

- Paper 101bis [5]: Strategy B′ replaces Strategy B (KS-adaptive radius), $\mathbb{E}[s^\star_{B'}] = O(1)$.
- Paper 101ter [6]: α-theoretic Armijo amortisation, $\mathbb{E}[j_{\mathrm{accept}}] = O(1)$.
- Paper 101 [4]: ABD refuted under backtracking-only Armijo, $\mathbb{E}[N_{\mathrm{pre}}] = \Theta(d^{0.84})$, REC stated.
- Paper 101quater (this paper): ABD rescued under extended line search, $\mathbb{E}[N_{\mathrm{pre}}] = O(1)$, REC superseded.

The remaining open work is on the analytic side: formal proofs of Plancherel decorrelation (Lemma 3.2 of paper 101bis) and of the optimal-τ scaling (Lemma 3.1 of this paper). The empirical evidence for both is overwhelming.

Recommendation. In all future work in this series, replace the Armijo backtracking of Part VII Definition 2.1 by the extended line search of Definition 2.1. The change is two lines of code, the speedup is $\Omega(d/\log d)$ on KS.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [7] or Lairez [9].

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